

Summary of methodological appraisal and reporting limitations

Risk of bias was appraised by study design; unusually high performance estimates were treated as interpretive cautions, not integrity determinations.

Experimental studies (animal components, n = 7)

High risk of bias  3

Predominantly unclear reporting  4

Predominantly unclear reporting was driven by missing information on allocation concealment, random housing, and blinded assessment.

Human observational studies (n = 5)

NOS 7/9 (lowest risk)  1

NOS 6/9  3

High risk on QUIPS  1

The prognostic-factor cohort was rated high risk on QUIPS because the sample was small relative to the number of candidate predictors.

Interpretive cautions (not formal risk-of-bias domains)

Two reports yielded unusually high diagnostic or prognostic performance estimates. These findings were interpreted cautiously because of potential spectrum bias, small samples, and risk of overfitting.